

Shark Movement Forecasting: An Assessment for Ocean Recreational Safety

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Synopsis

Shark presence is an important but difficult-to-predict factor affecting ocean recreation safety. This project examines whether tiger shark (*Galeocerdo cuvier*) presence near the eastern coast of Cape Canaveral, Florida, can be predicted using a combination of environmental, spatial, and temporal conditions. By integrating logged shark detections with offshore buoy historical meteorological data spanning from 2010-2019, several supervised and unsupervised models were evaluated to understand when certain factors align with elevated shark presence risk. The results show that spatial location is a dominant predictor, while environmental and temporal variables contribute more limited predictive power on their own. These findings suggest that while predictive tools can help with identifying patterns of elevated shark presence risk, future improvements will require denser shark tracking data and more ecologically informative predictors before such models can be deployed in more interactive ways for casual public use as directionally informative tools to better understand risk indications for shark presence.

Introduction

Question

Can we predict the likelihood of tiger shark (*Galeocerdo cuvier*) presence along the western North Atlantic coastline near the East of Cape Canaveral, Florida, using spatial, temporal, and environmental conditions, in order to inform ocean recreation risk?

Background

As surfers and other people interested in using the ocean for recreation, safety is influenced by a wide range of environmental factors. Conditions such as wave height, swell direction, bacteria levels, and rip current risk are routinely monitored and predicted days in advance. In contrast, shark presence- while highly relevant to public safety - remains largely reactive, relying on real-time detection and eye-witness reporting. Although there are existing public platforms that track tagged sharks, few efforts have been made that attempt to forecast periods of amplified shark presence risk using environmental conditions.

Recent research suggests that this gap is becoming increasingly important to address. With Global Warming affecting historical water temperatures by increasing them in geographic locations further north, recent studies have seen migration patterns and ranges of Tiger sharks changing. A fairly recent study conducted by a team from Atlantic Shark Expeditions notes that *“Dr. Hammerschlag’s research revealed that tiger shark migration was happening both earlier in the year and extending farther north than previously documented. These changes were closely tied to rising sea surface temperatures, particularly in the waters off New England and Nova Scotia. The study found that for every 1 °C increase in sea surface temperature, tiger sharks shifted their range approximately 440 kilometers northward (‘[Tiger Shark Migration Shifting as Oceans Warm](https://atlanticsharkexp.com/tiger-shark-migration-shifting-as-oceans-warm/)’¹).”* These findings emphasize the growing relevance of exploring environmental conditions in furthering our understanding of shark distribution patterns.

¹ “Tiger Shark Migration Shifting as Oceans Warm - Atlantic Shark Expeditions.” *Atlantic Shark Expeditions*, 10 Apr. 2025, atlanticsharkexp.com/tiger-shark-migration-shifting-as-oceans-warm/. Accessed 20 Dec. 2025.

It bears keeping in mind that tiger sharks are complex biological creatures that have mannerisms influenced by age, gender, life cycle events, and even their prey's movement. Another recent study focusing on movement patterns for tiger sharks in the Gulf of Mexico ended up resulting in some of the following conclusions: *“Our analyses revealed significant ontogenetic and seasonal differences in distribution patterns as well as across-shelf (i.e., regional) and sex-linked variability in movement rates. Presumably sub-adult and adult sharks achieved significantly higher movement rates and used off-shelf deeper habitats at greater proportions than juvenile sharks, particularly during the fall and winter seasons. Further, female maximum rate of movement was higher than males when accounting for size (‘[Movement patterns and habitat use of tiger sharks \(Galeocerdo cuvier\) across ontogeny in the Gulf of Mexico](#)’²).*” These biological drivers cannot be directly observed in most publicly accessible shark tracking datasets thus this research project focuses on probabilistic risk estimation combining available tiger shark movement tracking data with environmental and temporal conditions data to explore if statistical models will be able to identify periods connected to elevated risk of coastal shark presence using a binary classification system (shark presence vs shark absence), providing actionable insights for surfers and ocean recreation stakeholders.

Hypothesis

While tiger shark movement is influenced by biological factors such as life stage, and prey availability, we hypothesize that environmental and temporal conditions - particularly

² Ajemian, Matthew J et al. “Movement patterns and habitat use of tiger sharks (*Galeocerdo cuvier*) across ontogeny in the Gulf of Mexico.” PloS one vol. 15,7 e0234868. 15 Jul. 2020, doi:10.1371/journal.pone.0234868, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7363083/> . Accessed 20 Dec. 2025.

temperature, sea level pressure, wind conditions, and seasonality - are associated with elevated likelihood of coastal presence and can be anticipated using a predictive risk model.

Prediction

We expect that integrating shark movement data with environmental and temporal conditions will produce a model capable of identifying periods of heightened shark presence risk, particularly during warmer temperatures, calmer weather conditions, and known seasonal migration windows.

Data

Data Acquisition

This project integrates tiger shark movement tracking data with coastal environmental measurements to evaluate whether shark presence near the coastline can be predicted using environmental and temporal conditions.

The tiger shark movement dataset was created by the University of Miami as part of their project titled “[Regularized satellite tracks from: Ocean warming alters the distributional range, migratory timing, and spatial protections of an apex predator, the tiger shark \(*Galeocerdo cuvier*\)](https://doi.org/10.1111/gcb.16045)³” and is publicly available online as a comma-separated file (.csv).. This dataset contains

³ Hammerschlag, Neil, et al. “Ocean Warming Alters the Distributional Range, Migratory Timing, and Spatial Protections of an Apex Predator, the Tiger Shark (*Galeocerdo Cuvier*).” *Global Change Biology*, vol. 28, no. 6, Jan. 2022, <https://doi.org/10.1111/gcb.16045>

time-stamped latitude and longitude detection positions for tagged tiger sharks from 2010 to 2019, providing spatial and temporal observations of confirmed shark presence.

Environmental and weather data were sourced from the National Data Buoy Center (NDBC) and specifically from [Buoy 41009](#)⁴, which is based 20 nautical miles (NM) East of Cape Canaveral, FL, and has further details as shown in Figure 1 below:

Station 41009 ([LLNR 840](#)) - CANAVERAL 20 NM East of Cape Canaveral, FL

Owned and maintained by National Data Buoy Center

3-meter discus buoy

SCOOP payload

28.508 N 80.185 W (28°30'27" N 80°11'6" W)

Site elevation: sea level

Air temp height: 3.4 m above site elevation

Anemometer height: 3.8 m above site elevation

Barometer elevation: 2.4 m above mean sea level

Sea temp depth: 2 m below water line

Water depth: 42 m

Watch circle radius: 115 yards

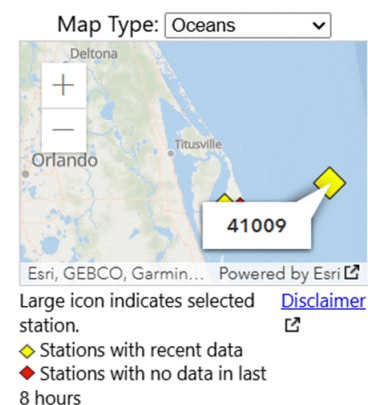


Figure 1: Showing additional information pertaining to Buoy 41009 as pulled from the National Data Buoy Center

This buoy was chosen due to its proximity to the study region and its overlap with areas of frequent shark detections as seen in the tiger shark movement tracking dataset. Historical standard meteorological data was pulled by year from 2010 to 2019, to match the coverage period of the tiger shark movement dataset years, as text files.

These datasets are well-suited to the research question for two reasons. First, the shark tracking dataset provides verified shark detections with timestamps and locations. Secondly, the

⁴ National Data Buoy Center maintained by the National Oceanic and Atmospheric Administration of the US Government. "NDBC - Station 41009 Recent Data." *Noaa-Mirror.org*, 2025, noaa-mirror.org/www.ndbc.noaa.gov/station_page.php@station=41009

buoy data provides environmental conditions measured consistently over the same period. Combining these sources allows the model to learn both when sharks were detected and the ocean and atmospheric conditions when that occurred.

The **response variable** is a binary classification indicator of shark presence (named “presence” within the final dataset), where the value of **1** represents a confirmed shark detection (presence) at a given time and location while the value of **0** represents an absence of a shark detection (absence). The absences were created using time periods in which there were no shark detections but environmental buoy data was available. This set-up will allow for the identification of the co-occurring environmental conditions associated with periods of elevated shark presence risk for ocean recreation users. It should be noted that this shark tracking dataset only includes tagged sharks, which may not be fully representative of the broader tiger shark population and that detection frequency may be influenced by tag performance, shark behavior, and receiver coverage, meaning that absences cannot guarantee true biological absence of sharks. Additionally, the buoy measurements are coming from the fixed location of the buoy located offshore and so it may not perfectly capture nearshore or more distant nuanced micro-conditions. These limitations are important considerations to keep in mind when it comes to interpretation of model results.

See the Appendix for a complete data dictionary of the final cleaned modeling dataset (“final_model_dataset_shark_presence.csv”).

Cleaning

Initial exploration of the raw datasets revealed substantial missingness that would need to be resolved, and variables with high correlation that would likely need to be dropped to avoid issues with multicollinearity. It also showed the need for developing a complete presence-absence variable for the sharks to use the logged shark detections and non-detections to further project analyses. The shark movement and buoy datasets were cleaned separately before being merged into a single modeling data set. Data cleaning and preparation were conducted in Python using reproducible, well-documented code to ensure consistency and allow for replication by other analysts.

For the Buoy 41009 data, the separated annual meteorological text files, which covered from 2010-2019, were combined into a single comma-separated values (csv) formatted dataset. In reviewing the [National Data Buoy Center's Measurement Descriptions and Units](#)⁵ section, which outlines the abbreviations and data found in the datasets, it noted that “MM” and variable number of 9’s are used to denote missing data. So in an effort to prevent any inadvertent misinterpretation of those integers in statistical analysis, missing data was switched to reflect NaN. The initial review revealed completely empty variable columns, including station visibility (VIS) and the water level in feet above or below Mean Lower Low Water (TIDE), so those were removed as they had no information to offer. Variables were retained only if they were consistently measured over the years and were determined to be plausibly relevant to shark movement and/or ocean recreation risk. The below Figure 2 contains the correlation heatmap for the Buoy 41009 dataset to be able to compare the relationship between variables:

⁵ US Department of Commerce, National Oceanic and Atmospheric Administration. “National Data Buoy Center - Measurement Descriptions and Units.” *W*www.ndbc.noaa.gov, www.ndbc.noaa.gov/faq/measdes.shtml

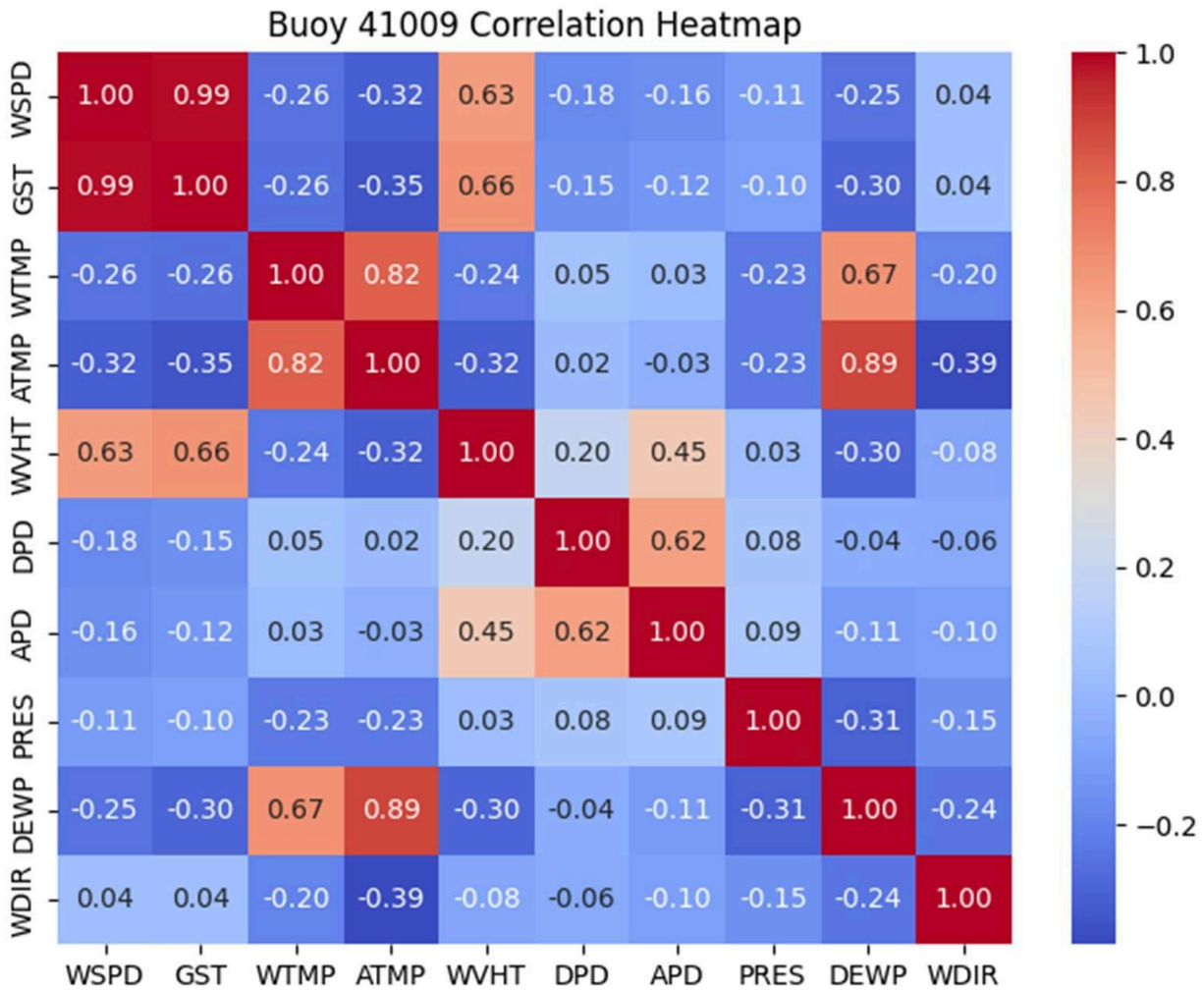


Figure 2: Correlation heatmap for continuous Buoy 41009 variables

Figure 2 shows an extremely high correlation (about 0.99 of 1.0) between peak 5 or 8 second gust speed (GST) and windspeed (WSPD). Strong correlations were also observed between air temperature (ATMP) and both dew point (DEWP) and water temperature (WTMP). The level of missingness was compared in each highly correlated pair with the less complete variable being removed and so, the following variables were dropped at this point: peak 5 or 8 second gust speed (GST), Dew Point (DEWP), and Water temperature (WTMP). The missingness report also showed that there were four variables that were over 50% incomplete: Mean Wave Direction

(MWD) at about 77%, while Dominant Wave Period (DPD), Wave Height (WVHT), and Average Wave Period (APD) were all about 59% incomplete. These variables were removed since the remaining variables were showing only about 2% or less missingness and trying to impute such high amounts of missing data may add noise and detract from available actual environmental data.

For the tiger shark movement tracking dataset, timestamps were standardized and converted to be a consistent datetime format while only valid latitude-longitude shark detections associated with timestamps were retained. Because the shark tracking dataset records were already known to only contain confirmed detections of individual tagged sharks, all of the remaining observations in this dataset were treated as true shark presence events.

The two datasets were merged using temporal alignment with shark detection timestamps being matched to the closest available timestamp available in the buoy dataset. Following the joining of the two datasets, the **binary response variable (presence)** was created, where the value of **1** indicates a confirmed shark detection (presence) and the value of **0** indicates no shark detection (absence) during the periods when environmental data was available. Absence observations were restricted to the time periods found to be active shark detection windows to reduce temporal bias. To further address class imbalance while maintaining the temporal structure of the dataset, absence observations within those active windows were downsampled to maintain an approximate 3:1 absence-to-presence ratio. The intention of this approach was to reduce model bias towards the majority class (absence) without falsely inflating sample size or disrupting any season patterns.

It is important to keep in mind that some assumptions underlie these data cleaning decisions such as the presumption that absence observations represent non-detections rather than confirmed biological absence of sharks since only tagged sharks were being monitored for detection observations. Secondly, buoy measurements were read from a fixed offshore location and are being assumed to approximate broader coastal environmental conditions relevant to shark presence in the project focus area. These assumptions create potential threats to validity and are revisited within the Discussion section. The final cleaned dataset contains approximately twenty-thousand (20,000) observations, the target presence variable, and thirteen (13) predictor variables, which include a mix of environmental, temporal, and spatial features. A complete data dictionary for this final cleaned dataset is provided in the Appendix.

Data Exploration

Exploratory data analysis (EDA) was conducted on the final cleaned dataset to understand environmental variability, class balance, temporal patterns, and relationships among predictors prior to model fitting to understand the best practices to avoid data leakage or other unintentional issues based on handling of the dataset when setting up modeling pipelines. All figures and tables referenced below in this section were created using the cleaned dataset described in the previous section.

Figure 3 below shows that the shark detections are clustered in specific time periods within the modeling window. Shark presence is not evenly distributed across time, with distinct points of elevated activity scattered among long stretches of non-detection. This temporal clustering reinforces the ecological reality that shark presence is episodic rather than continuous

and supports the decision to restrict absence observations to active detection windows rather than treating all periods of time equally.

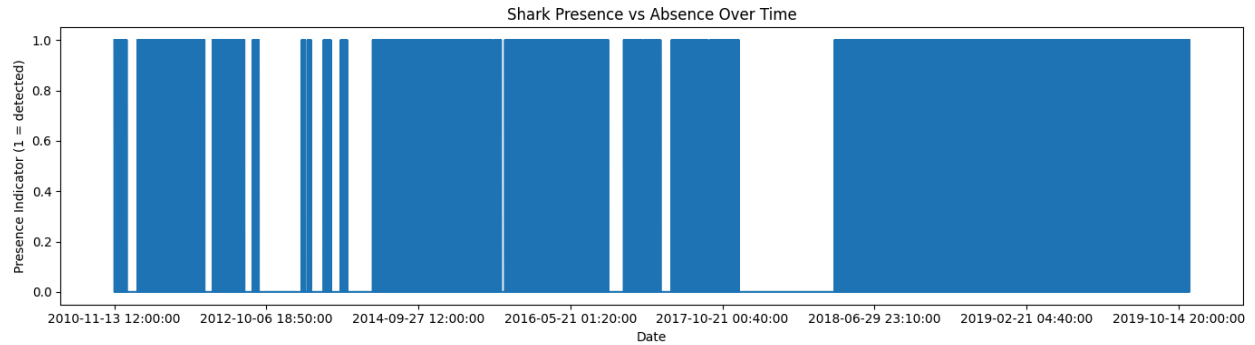


Figure 3: Shark presence indicator over time, aggregated by timestamp

To address class imbalance while preserving temporal structure, absence observations were downsampled to an approximate 3:1 ratio relative to the presence observations. Table 1 below summarizes the resulting class distribution and while absence remains the majority class, the reduced imbalance serves to improve model learning without artificially inflating the number of presence detections in the dataset.

Class Balance Table		
Presence	Count	Percent
0	15546	75
1	5182	25

Table 1: Class Balance Table (Presence vs Absence %) of the final modeling dataset after downsampling

To evaluate any potential multicollinearity, a correlation heatmap of the continuous predictor variables for the final clean dataset was created (see Figure 4). Previous rounds of data cleaning removed meteorological variables showing strong correlations and any variables showing high missingness to reduce redundancy. Most importantly, no single retained variable appeared to be perfectly correlated with the presence response variable, which suggests that no sole predictor determines shark presence.

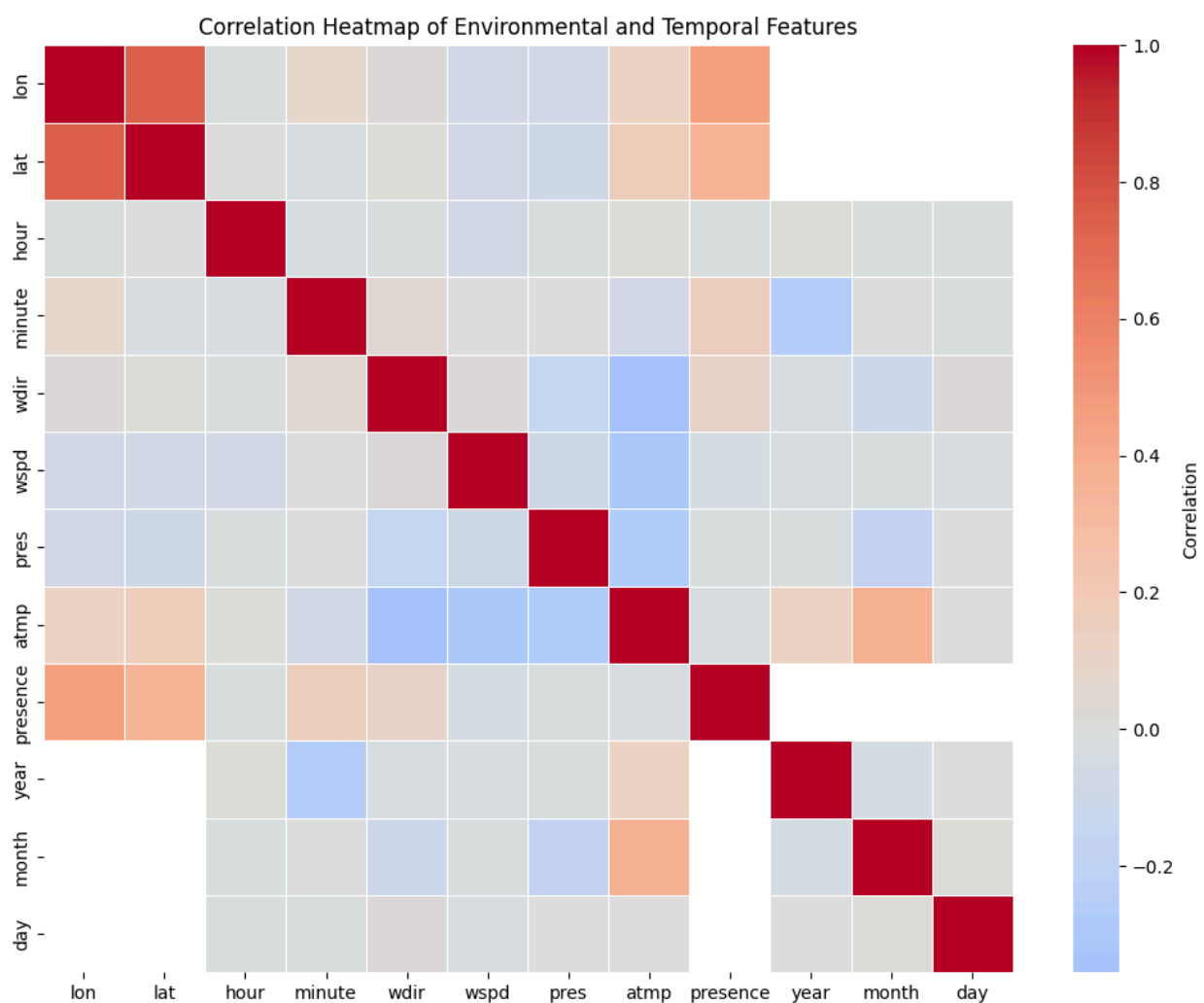


Figure 4: Correlation Heatmap for Final Cleaned Dataset

Warmer air temperatures appear to be associated with shark presence (see Figure 5). Though because of the substantial overlap between presence and absence air temperature conditions, it supports a multivariate modeling approach as it indicates that temperature alone will likely be insufficient for reliable detection predictions.

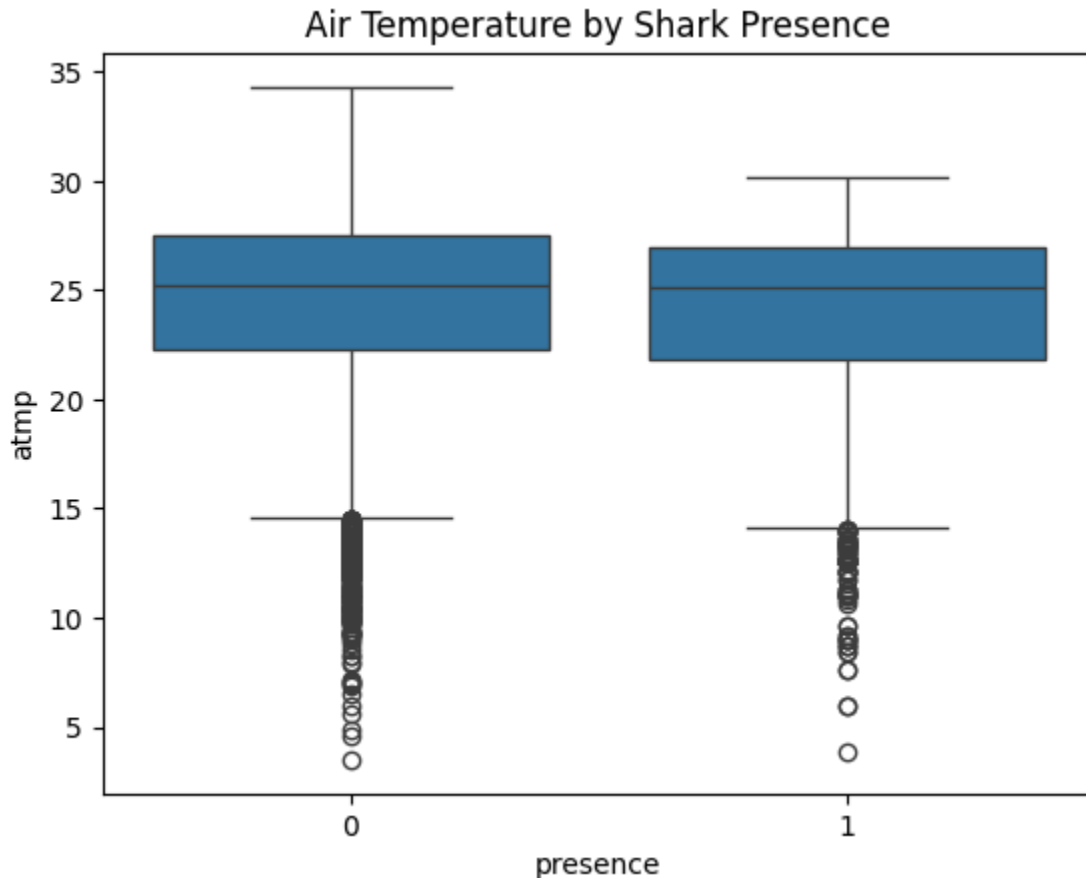


Figure 5: Boxplot for Air Temperature by Shark Presence

Wind speed showed similar medians between presence and absence with presence appearing to have the narrower distribution of the two (see Figure 6). It may be the case that extreme wind

conditions may reduce detectability and/or shark activity. Wind speed is likely more of a contextual variable than a driving variable when it comes to presence.

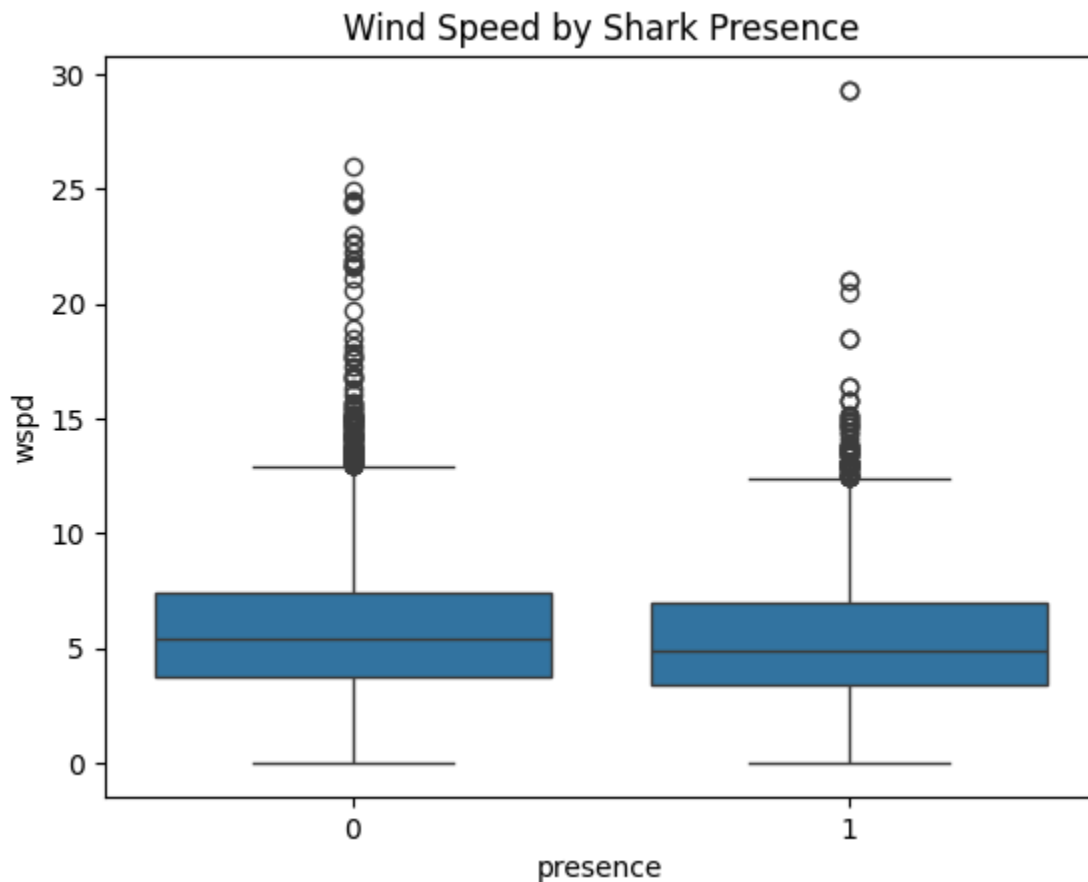


Figure 6: Boxplot for Windspeed by Shark Presence

Sea-level pressure clusters in the boxplot pulled (see Figure 7 below) were tightly around the typical high sea-pressure conditions, which for reference “regular sea-level pressure” is defined as 1013.25 hPa also known as 1 atmosphere, with fewer extreme low-pressure values during presence detections. Shark presence may be less likely during storm systems as most detections appear to have been during stable conditions.

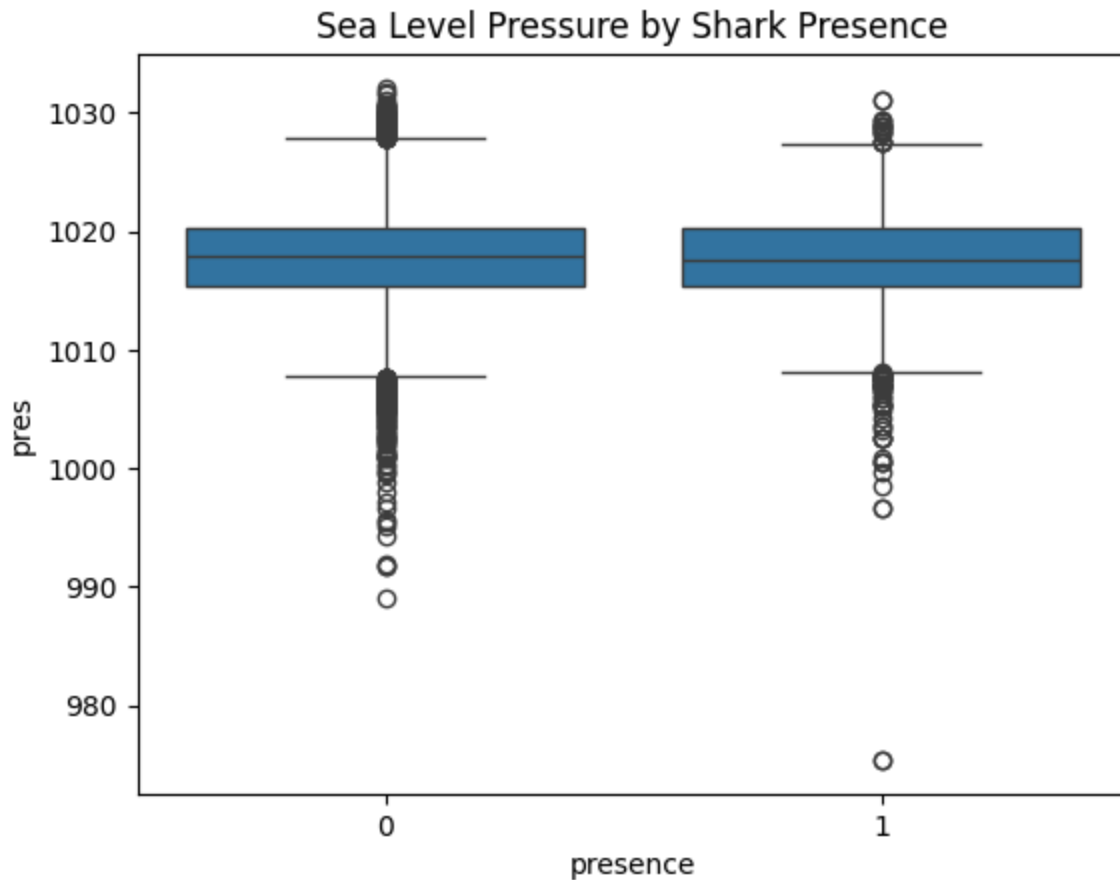


Figure 7: Boxplot for Sea Level Pressure by Shark Presence

The distributions of key environmental variables were explored to assess their variability. Figure 8 displays the distributions of wind speed (m/s), air temperature (degrees in Celcius), sea-level pressure (hPa), and hour of day within the 100% shark presence cluster created when using Hierarchical Clustering. Shark presence detections tend to occur within moderate wind speeds, warmer air temperatures, calm sea-level pressure (as a reminder, regular sea-level pressure is defined as 1013.25 hPa also known as 1 atmosphere), and during late morning hours.

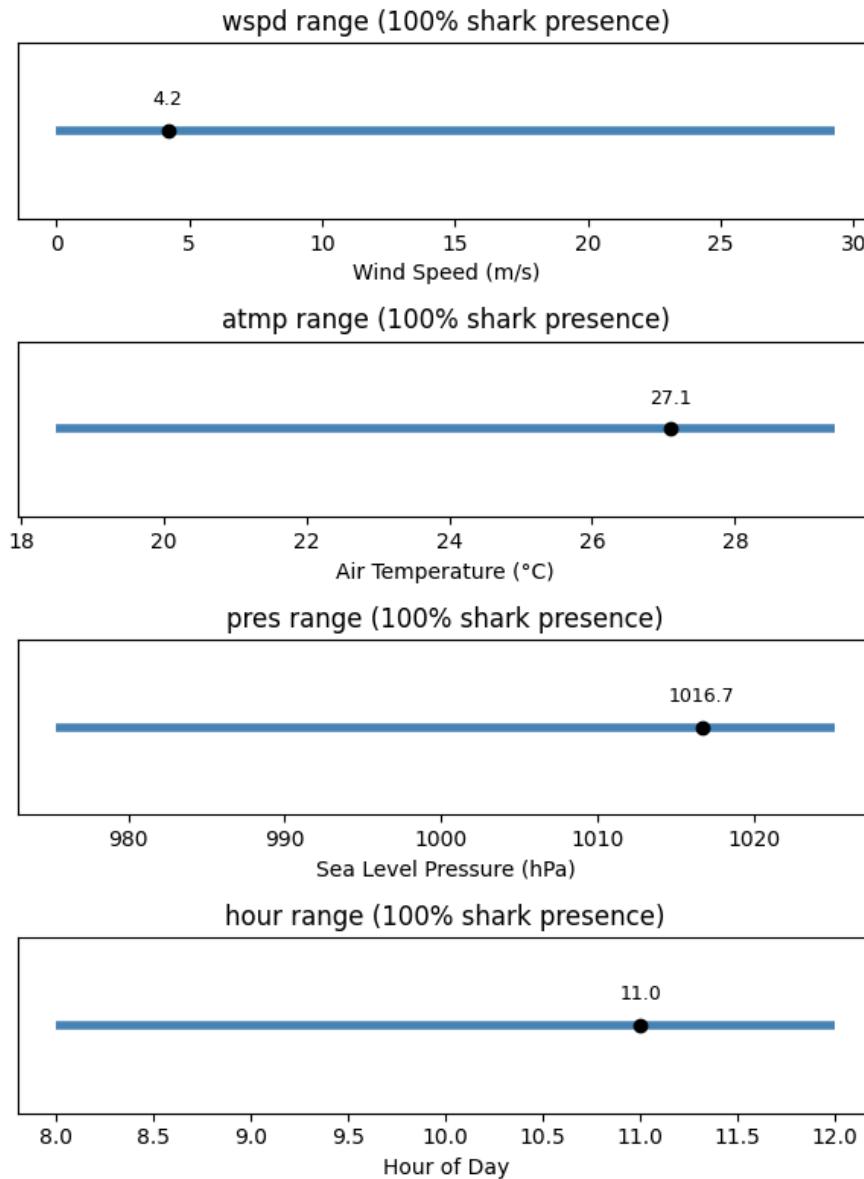


Figure 8: Cluster-Level Environmental Ranges for the 100% shark presence cluster created using Hierarchical Clustering

Overall, the exploratory analysis revealed quite a bit about our dataset. It showed that shark presence is temporally clustered rather than uniformly distributed over time. Additionally,

environmental variables were showing realistic ranges and meaningful variability. Moderate correlations had existed among certain predictors on previous versions of the dataset, but multicollinearity was mitigated through feature selection so our final dataset shouldn't have issues with that. These findings helped to inform modeling choices, feature selection, and evaluation strategies described in subsequent sections.

Models

Preprocessing and dimensionality reduction - *feature selection/engineering*

Prior to fitting supervised predictive models, several preprocessing and feature preparation steps were utilized to ensure that the modeling inputs were appropriately structured, comparable in scale, and informative for downstream analysis. These steps were designed to improve model performance and reduce risks of overfitting, data leakage, and time leakage. Separate preprocessing pipelines were set up for the unsupervised modeling and supervised modeling due to their different requirements to run. Spatial coordinates (latitude and longitude) were initially included to allow models to learn geographic structure but their overall impact was later revisited through spatially restricted models to assess generalizability of models in new locations.

All continuous predictors were standardized prior to modeling to ensure that variables initially measured on differing scales contributed comparably during optimization. The response variable was already set up for binary (shark presence vs. absence) classification so no additional transformations were needed for it. A time-aware stratified train-test split was used to preserve class balance while preventing chronological leakage in this temporally structured ecological

dataset as the split attempts to approximate real-world forecasting conditions where only past information would be available for predictions on unseen future conditions.

Unsupervised learning techniques were used as exploratory steps to better understand the structure of environmental conditions associated with shark detections. Principal Component Analysis (PCA) was applied to assess whether variance in standardized environmental variables was controlled by a small number of factors, however, the results showed that no single factor was dominating them (Figure 9). Additionally, different clustering techniques like K-Means Clustering, Hierarchical Clustering, and Gaussian Mixture Modeling, were used to identify common environmental and temporal trends. Although the unsupervised models results were not directly used as predictive features in the final supervised models, they helped inform interpretation by highlighting condition characteristics associated with elevated shark presence and so inspired later analyses examining environmental and temporal ranges for high-risk shark presence periods.

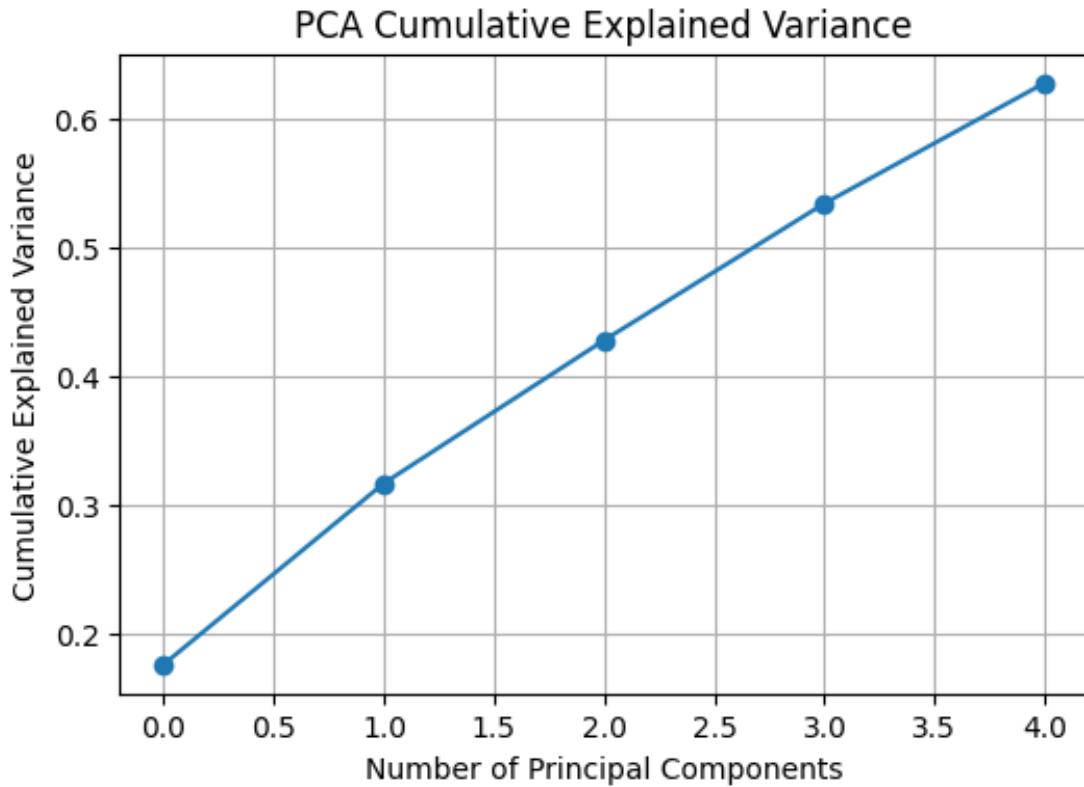


Figure 9: Gradual variance increases as additional components are added

Overall, the preprocessing and dimensionality reduction steps taken were designed to balance predictive performance, interpretability, maintain temporal and data integrity by avoiding leakage, and statistical validity. By standardizing predictors, preserving temporal structure, and introducing the use of unsupervised models for exploratory insight, the modeling pipeline was set up to support robust inference while furthering the project's predictive goals. These steps laid the groundwork for selecting and evaluating supervised classification algorithms, which are discussed in the next section.

Algorithms Selection

To address the research question of whether tiger shark presence can be predicted from spatial, temporal, and environmental conditions, multiple supervised learning algorithms were explored using a binary classification framework. Algorithm choices were influenced by interpretability, robustness to multicollinearity, suitability for ecological relationships, and relevance to the project's overarching goal of being capable of real-world risk communication to surfers and ocean recreation stakeholders.

Logistic Regression was selected to be used as the baseline model due to its general ease of interpretability and strong foundation for use in binary classification tasks. It provides a transparent relationship between predictor variables and the probability of shark presence, which is beneficial in allowing for stakeholders to more easily understand which conditions most strongly influence predicted shark presence risk. Both a **baseline “unpenalized” Logistic Regression model**, which uses the default L2 regularization with no tuning, and **“penalized” Logistic Regression models**, which used the following regulation of L1 - Lasso (feature selection pressure), L2 - Ridge (stabilizes correlated predictors), and ElasticNet (which is a hybrid of L1 & L2), were implemented to assess sensitivity to multicollinearity, to control potential overfitting, and to serve as valuable benchmark models against which more complex models can be evaluated.

Gradient Boosting Classification was selected as a non-linear modeling approach and is known for its ability to capture more complex relationships in tabular data with its capabilities of being relatively robust to mixed-scale predictors. This approach is particularly appropriate for working with ecological systems where shark movement is influenced by its interactions with

environmental and temporal factors. Cross-validated hyperparameter optimization was used to tune the models to try to balance predictive performance with generalization.

Random Forest Classification was selected as another non-linear modeling approach and its known abilities to perform well in settings with noisy predictors and robustness when it comes to correlated predictors. Including this modeling approach allowed for performance comparison between the non-linear models (Random Forest and Gradient Boosting) to validate whether observed predictive outputs were consistent across different non-linear based model frameworks.

All of the supervised models' performances were evaluated using ROC-AUC, precision-recall curves, and train-test comparisons to assess overfitting risk, classification accuracy, and minority-class performance. This allowed for a comprehensive assessment of the benefits of linear versus non-linear approaches and informed selection of a best-performing model appropriate for this current stage of this project.

Best Model

The best model selected for analysis at this stage of the project is the **Penalized Logistic Regression with ElasticNet regularization** after systematic performance comparison with the variety of logistic regression models utilized, gradient boosting classifier models, and random forest classifier models. The logistic regression models were selected as the primary modeling framework due to their interpretability, and suitability for binary classification. Hyperparameter tuning was conducted using five (5) fold cross-validation to optimize regulation strength and balance, with model performance evaluated using ROC-AUC as the primary metric to account for class imbalance.

```

=== PENALIZED LOGISTIC REGRESSION MODELS ===

--- L1 Logistic Regression ---
precision    recall    f1-score   support

     0      0.95      1.00      0.98      4353
     1      1.00      0.86      0.92      1451

 accuracy
macro avg      0.98      0.93      0.95      5804
weighted avg    0.97      0.96      0.96      5804

L1 ROC-AUC: 0.9422

--- L2 Logistic Regression ---
precision    recall    f1-score   support

     0      0.95      1.00      0.97      4353
     1      1.00      0.84      0.92      1451

 accuracy
macro avg      0.98      0.92      0.95      5804
weighted avg    0.96      0.96      0.96      5804

L2 ROC-AUC: 0.9419

--- ElasticNet Logistic Regression ---
precision    recall    f1-score   support

     0      0.95      1.00      0.98      4353
     1      1.00      0.85      0.92      1451

 accuracy
macro avg      0.98      0.92      0.95      5804
weighted avg    0.96      0.96      0.96      5804

ElasticNet ROC-AUC: 0.9422

Summary of ROC-AUC Scores:
L1: 0.9422
L2: 0.9419
ElasticNet: 0.9422

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Figure 10: Logistic regression models performance metrics outputs

Across all penalized logistic regression variants, model performance was strong and stable as shown in Figure 10. The ElasticNet model achieved a ROC-AUC score of 0.9422, which marginally outperformed the ridge regression and lasso models. The model was successful at identifying the majority of true shark presence while minimizing false alarms as suggested by its high precision of 1.00 and recall for presence remaining steady at 0.85. Although the tree-based non-linear models of gradient boosting and random forest managed to achieve

near-perfect classification metrics, these models were found to rely primarily on spatial features and exhibited signs of overfitting thus limiting their interpretability and generalizability. As such, the ElasticNet linear regression model was selected as the current best model, proving itself capable of balancing robustness, interpretability, and predictive performance.

To further assess the extent to which the model may be depending upon the spatial information, a spatially-restricted logistic regression model was evaluated with latitude and longitude removed (see Figure 11). The performance substantially dropped to a ROC-AUC score of 0.64 and the model failed to meaningfully detect shark presence observations thus emphasizing the strong influence of spatial structure in the data. This post hoc analysis demonstrates that while spatial features substantially improve prediction accuracy across all of the models, that spatial reliance also limits the ability for the model to be able to successfully be used in new locations. The ElasticNet model is able to identify elevated shark presence risk conditions within historically observed regions, rather than a model capable of being a fully generalizable shark forecasting system. Future improvements will seek to reduce model spatial dependence and improve temporal and environmental generalization.

=== SPATIALLY-RESTRICTED LOGISTIC REGRESSION RESULTS ===				
	precision	recall	f1-score	support
0	0.75	1.00	0.86	4353
1	0.00	0.00	0.00	1451
accuracy			0.75	5804
macro avg	0.38	0.50	0.43	5804
weighted avg	0.56	0.75	0.64	5804
ROC-AUC: 0.6391034297029403				

Figure 11: Performance metrics for the spatially-restricted logistic regression model

Discussion & Next Steps

This project set out to explore whether tiger shark (*Galeocerdo cuvier*) presence near the eastern coast of Cape Canaveral, Florida, could be predicted using a combination of spatial, temporal, and environmental data, with the overarching goal of creating a directionally informative ocean recreation risk assessment tool. Shark movement tracking data was integrated with buoy-based environmental measurements and said dataset was used in a series of supervised and unsupervised models that were evaluated to understand their predictive performance capabilities and underlying patterns associated with shark presence. Supervised models results indicate that while predictive accuracy can be high when spatial features are included, environmental and temporal variables by themselves only offer limited predictive power.

The strongest performing model at this stage in the project is a penalized logistic regression model that uses ElasticNet regularization and was able to achieve a ROC-AUC score of about 0.94. Further exploration revealed that longitude and latitude were dominating predictive importance in all of the models, thus raising concerns about overfitting and limited generalizability for the model to be used in new locations. A stress-test model with spatial variables removed was then developed and saw a substantial drop in performance when it only received a ROC-AUC score of about 0.64. This indicates that environmental and temporal conditions alone are insufficient when it comes to reliable shark presence prediction.

Unsupervised clustering methods identified consistent environmental and temporal ranges associated with periods of confirmed shark presence, including moderate wind speeds, warm air temperatures, typical sea-level pressure, and late- morning hours. These patterns of

environmental profiles that may elevate the likelihood of shark detections are best interpreted as risk context indicators, not guaranteed predictors of shark behavior or movement, which remains influenced by a complex array of biological factors such as age, sex, prey availability, and migration cycles.

Several limitations should be kept in mind when interpreting these results. First and foremost is that all statistical models are approximations so assumptions, whether implied or clearly stated, are never exactly true. Models should not be used for real-time safety guarantees, beach closure decisions, as a replacement for standard safety procedures nor are they meant to replace life guards, other official warning systems, or human judgement in any way. Models and findings should be viewed as tools to better understand risk indications for shark presence. Shark detections are limited to tagged sharks, meaning that absence observations represent non-detections rather than confirmed shark absence. Additionally, detection frequency may also be influenced by tag performance, receiver coverage, or shark behavior. It should also be noted that environmental conditions were derived from a buoy fixed offshore and as such, conditions might not capture all of the environmental condition nuances experienced closer to shore by surfers or ocean recreation individuals. Because of the models' strong dependence on spatial features, it raises concerns about model overfitting and potential transferability to new locations, future climate conditions, or shifts in shark migration behaviors.

In the short term, environmental cluster data could be used as risk context for surfers and current model output could be compared to existing shark monitoring systems to see how well they match up. Future work should prioritize expanding and improving the underlying data sources to incorporate more frequent spatially dense shark tracking data - potentially from additional tagging programs and/or shark species - as that would help address class imbalance

and improve model learning. Looking ahead to long term future goals, exploration of potential model integration into surf forecast-style presence risk indicators to provide directional awareness as to when certain conditions may warrant increased caution. Another long term goal would be to find opportunities for partnership with marine researchers, and expansion of tagging sharks efforts to further studies.

Code Availability

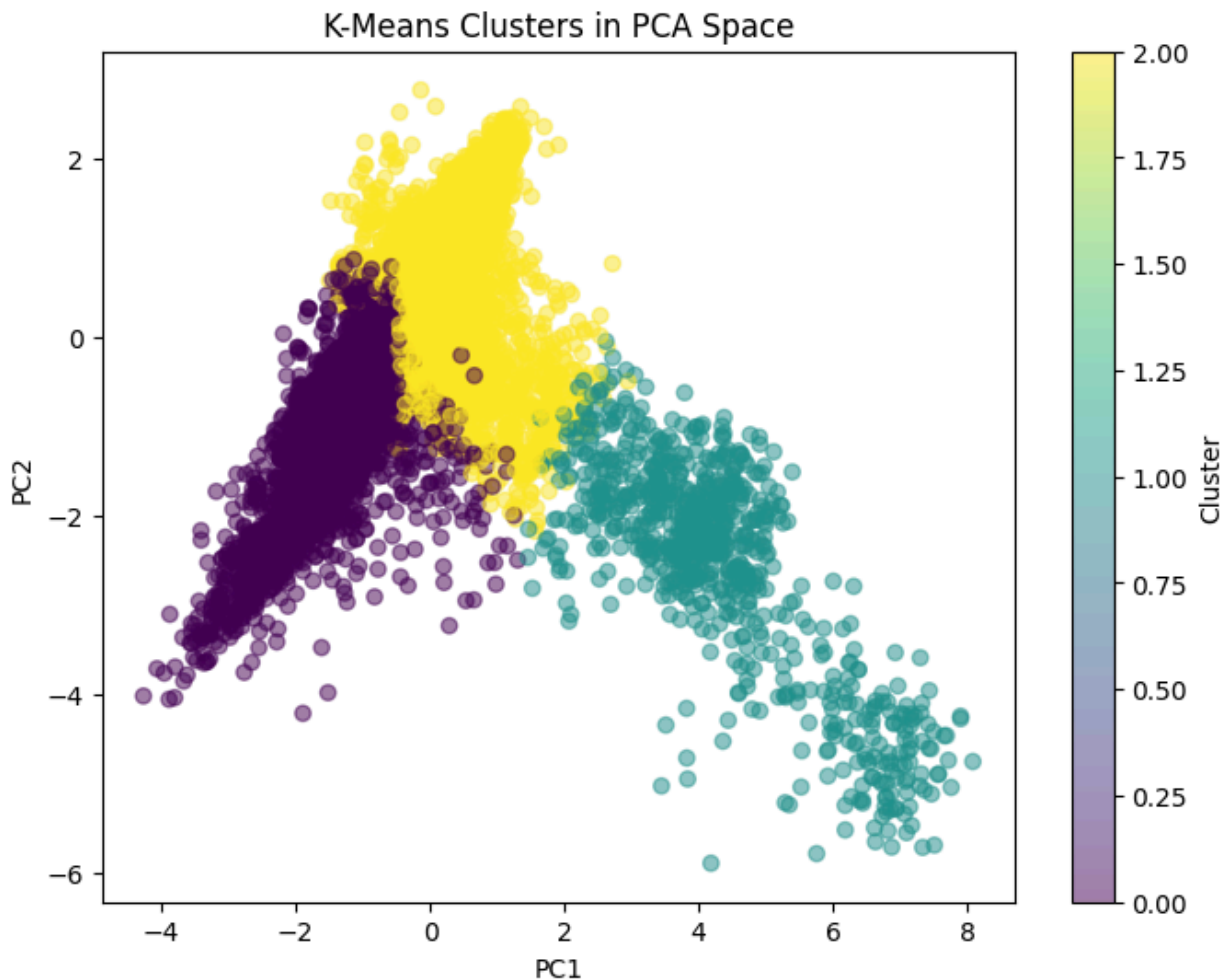
Github Repository: https://github.com/ParlatoI/Shark-Forecasting_Capstone/tree/main

Appendix

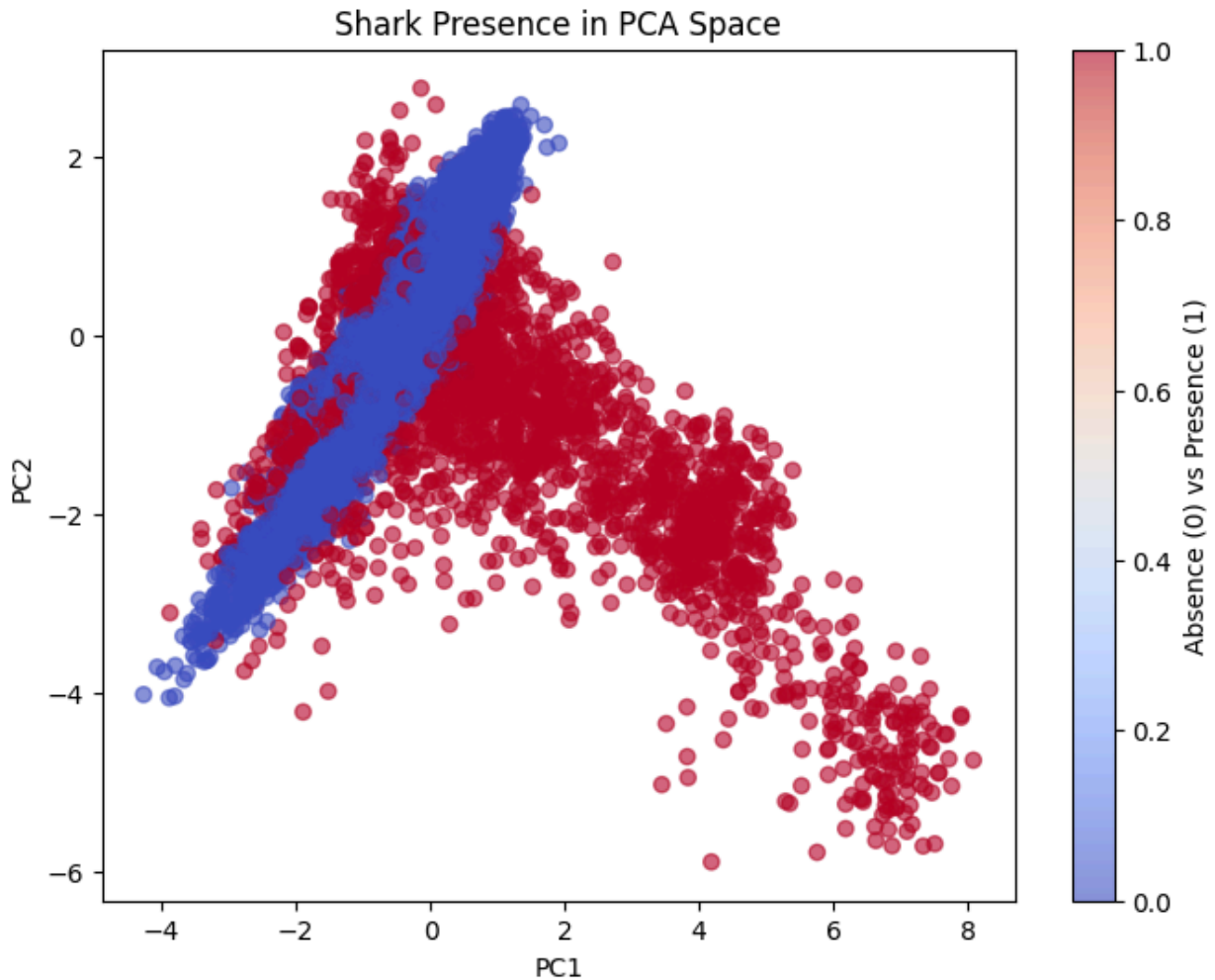
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lat	lat	float64	0.0000	28.508
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hour	hour	float64	0.0000	19.0
minute	minute	float64	0.0000	20.0
wdir	wdir	float64	0.0037	328.0
wspd	wspd	float64	0.0003	10.1
pres	pres	float64	0.0174	1013.1
atmp	atmp	float64	0.0062	17.1
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presence	presence	int64	0.0000	0

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month	month	float64	0.0201	1.0
day	day	float64	0.0201	1.0

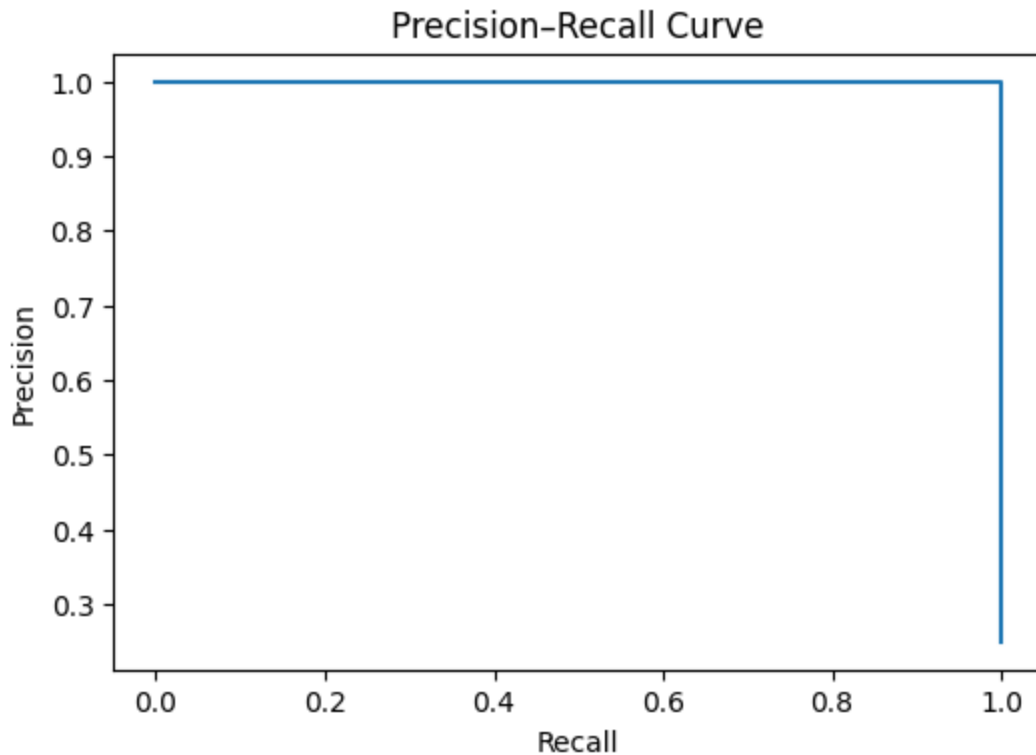
Appendix Table A1: Data Dictionary for final_model_dataset_shark_presence.csv



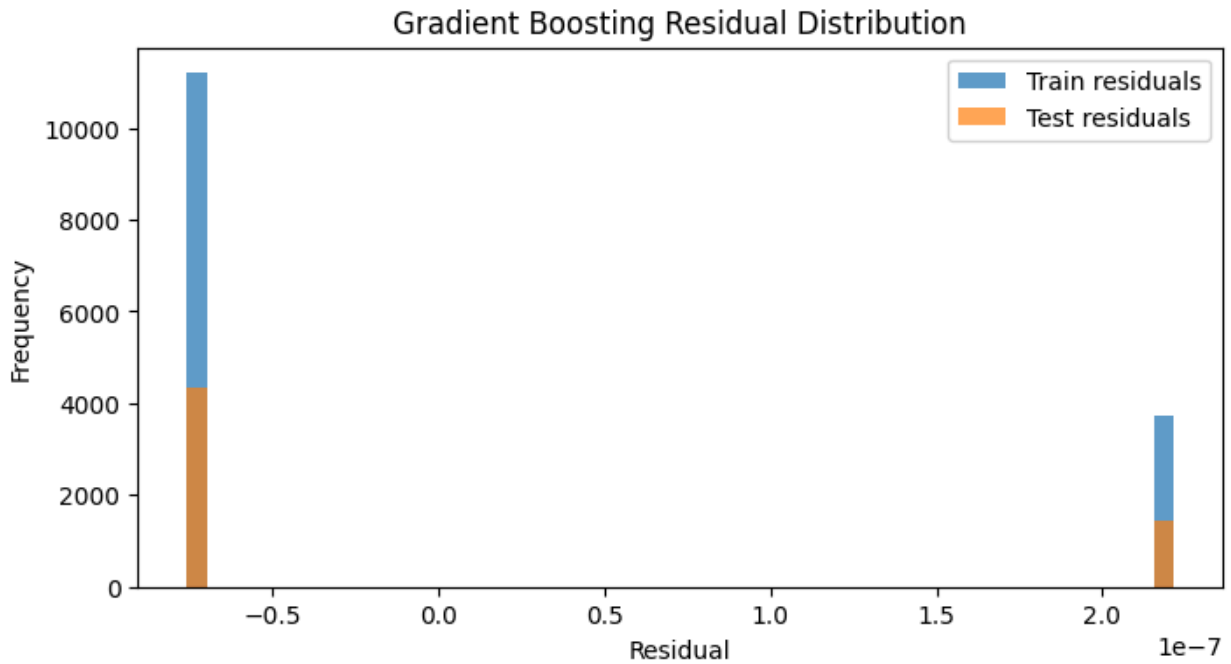
Appendix Figure A1: The three (3) K-Means clusters shown using PCA visualization appear distinct and well-separated thus indicating that K-Means identified meaningful structure for $k=3$ exploratory analysis.



Appendix Figure A2: For the K-Means clustering, this adds colors to shark presence vs absence showing that shark presence dominates the cluster 1 region. This visually confirms what we saw using crosstab (i.e. that shark detections appear under distinct conditions).



Appendix Figure A3: This is the precision-recall curve for the Penalized Logistic Regression model. The PR curve is too good at almost 1.0 precision for the entire recall range then sharp vertical drop at recall about 1.0. The model almost never predicts shark presence, incorrectly and when it does predict presence, it's almost always correct. The model is cautious and may miss some real shark events as it favors precision over recall.



Appendix Figure A4: This is the Residual Distribution plot for the initial Gradient Boosting Classifier model, which shows an extreme concentration near 0, and suggesting overconfidence and overfitting